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ORIGINAL RESEARCH

Epidemiology Landscape and Impact of Overweight and Obesity in Adults: Multi-country Results from the IMPACT-O Study

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ABSTRACT

Introduction: To estimate the recording rates and impact of overweight/obesity across selected countries in Europe and Asia–Pacific.
Methods: IMPACT-O was a retrospective cohort study of electronic medical records and claims databases from France, Germany, Italy, Spain, the UK, Australia, and Japan. Recording levels and the number of adults with overweight/obesity [based on diagnostic codes and/or body mass index (BMI)] during 2018–2022 were estimated. BMI thresholds/categories and obesity-related complications (ORCs) were described for adults with ≥ 1 BMI record of ≥ 25.0 kg/m^2 and ≥ 12 months observation before the index date.
Results: The number of active subjects across databases ranged from 1,203,674 to 22,413,902. BMI was recorded for 4.8–38.9% of active patients. The number of adults with overweight/obesity ranged from 3.5% in Australia to 24.9% in the UK. The percentage of individuals with BMI > 25 to < 30 kg/m^2 ranged from 43.6% in Spain to 77.9% in Japan; the percentage with BMI ≥ 30.0 kg/m^2 ranged from 22.1% in Japan to 54.8% in Spain. Among those with overweight/obesity, the percentage with diagnosis codes in their records ranged from 2.9% in France to 50.1% in Germany. Most (59.6–85.0%) individuals had ≥ 1 ORC; 35.9–65.8% had

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multimorbidity. The most frequently recorded ORCs were hypertension, dyslipidemia, depression, and type 2 diabetes.
Conclusion: Only a small proportion of people with overweight/obesity are formally diagnosed in healthcare databases. Most present with ≥ 1 ORC, commonly hypertension, dyslipidemia, depression, and type 2 diabetes.

PLAIN LANGUAGE SUMMARY

This study evaluated databases from seven countries (France, Germany, Italy, Spain, the UK, Australia, and Japan) to understand how individuals with overweight or obesity are recognized and recorded, and what health issues they face. The study looked at data from 2018 through 2022, focusing on adults who had at least one body mass index (BMI) measurement of 25 kg/m^2 or higher and a year of medical history on file. The study found that BMI was recorded for only 5–39% of patients, depending on the country. Among those with a BMI record, the proportion of adults classified with overweight or obesity varied from 3.5% in Australia to 25% in the UK. Of these, the majority were overweight (BMI between 25 and 30 kg/m^2), with proportions ranging from 44% in Spain to 78% in Japan. The rest had a BMI of 30 kg/m^2 or higher, from 22% in Japan to 55% in Spain. Only a small proportion of people with overweight/obesity had a coded diagnosis in their healthcare records. Most people (between 60% and 85%) with overweight or obesity also had at least one related health complication, and many had two or more complications. The most common problems were high blood pressure, high cholesterol levels, depression, and type 2 diabetes. Diagnosing obesity is the first step towards effective treatment, as under-recording of obesity in databases may lead to an underestimation of its impact, lower quality of care, and missed opportunities to provide early and effective medical support for obesity management.

Keywords: Obesity; Overweight; Databases; Epidemiology; Cardiovascular disease

Key Summary Points
Why carry out this study?
Obesity is a chronic progressive relapsing disease, which is reaching pandemic proportions across the world, and effective strategies to prevent and manage obesity are necessary.
Obesity is associated with a large number of health complications, which create significant clinical and economic pressure on healthcare systems across the world.
It is unclear if overweight and obesity are adequately recognized in healthcare databases.
What was learned from the study?
Recording of obesity (body mass index and/or diagnosis codes) by healthcare providers is insufficient, as revealed by analysis of major healthcare databases in seven countries in Europe and Asia–Pacific.
Most people in databases with records of obesity have at least one obesity-related complication, with the most frequent being hypertension, dyslipidemia, depression, and type 2 diabetes.
Strengthening efforts to increase awareness of overweight and obesity and recording levels among healthcare providers is warranted, as under-recording of obesity in databases may lead to an underestimation of its impact, lower quality of care, and missed opportunities to provide early and effective medical support for obesity management.

INTRODUCTION

Obesity is a chronic progressive relapsing disease affecting about one-quarter of adults in developed countries. According to data from the World Health Organization (WHO), in 2022, approximately 2.5 billion adults (43% of the world's population) presented with a body mass index (BMI) ≥ 25 kg/m^2 [1, 2]. Of these, 890

million (16%) were living with obesity [1, 2]. The prevalence of obesity has tripled since 1975, and is expected to increase further in the future [1, 3]. In Europe, overweight and obesity affect almost 60% of adults and almost a quarter (23%) are living with obesity [4]. From a public health perspective, obesity places significant clinical and economic pressure on the healthcare systems across the world, which is likely to intensify in the future [5, 6]. Although the prevalence of obesity varies across different populations, it is more common among those with lower socioeconomic status [7].

There is a clear and urgent need for effective strategies to prevent obesity, preferably beginning at a young age, and to curb the adverse health outcomes linked to obesity globally. The most common obesity-related complications (ORCs) include cardiovascular disease (CVD), type 2 diabetes (T2D), musculoskeletal disorders, and many cancers [3]. It is estimated that nearly 60% of all new diabetes cases will be due to obesity, as well as 18%, 11%, and 8% of all cases of CVD, dementia, and cancers, respectively [8]. A meta-analysis found consistent data from across the world indicating that both overweight and obesity are associated with higher all-cause mortality [9]. Additionally, a link between obesity, reduced health-related quality of life (HRQoL), and self-esteem has been established [10]. Obesity management, which includes lifestyle interventions, medications, or surgery, has been shown to improve cardiometabolic risk factors, ORCs, and cardiovascular mortality [11]. Several anti-obesity medications are also available with variable success rates and safety profiles [12].

One of the major hurdles to effective obesity management is the need for wider clinical recognition of obesity as a chronic disease at the population level, as well as the screening, diagnosis, and consistent recording in patient medical records [13]. The rate of BMI measurement and obesity diagnosis recording may help inform the extent of obesity recognition and formal diagnosis in healthcare settings. In addition, as new therapies become available to treat obesity, there is also a need to further understand and generate updated evidence on the epidemiologic landscape of overweight and obesity, and current management in adults at the population level [14]. Data routinely recorded by physicians in healthcare databases, including electronic medical records (EMRs) and claims databases, can help in understanding the real-world landscape and the impact of overweight and obesity. However, published epidemiologic data are limited.

Here, we report results from the epIdeMiology landscape and PAtient Care paThways of Obesity (IMPACT-O) study, a multinational real-world study, which used existing healthcare databases to characterize adults with overweight or obesity in seven European and Asia–Pacific countries: France, Germany, Italy, Spain, the United Kingdom (UK), Australia, and Japan. The IMPACT-O study evaluated the extent of BMI recording in healthcare databases in these countries, as well as the use of standard diagnosis codes for overweight and obesity by physicians. Further, the study described the profile of people with obesity or overweight identified in the databases including the number of ORCs that were reported.

METHODS

The IMPACT-O study was a multi-country, retrospective cohort study that utilized EMRs and claims databases standardized to the Observational Medical Outcomes Partnership (OMOP) Common Data Model (CDM). The OMOP CDM is developed and maintained by the Observational Health Data Sciences and Informatics initiative (https://ohdsi.github.io/CommonDataModel/). All analyses performed in this study were conducted in accordance with Data Use Agreement terms as specified by the data owners. For the databases for France, Germany, Italy, the UK, Australia, and Japan, there was no Institutional Review Board applicable to the usage and dissemination of the results of this study or required registration of the protocol with additional ethics oversight. For the database for Spain [the longitudinal patient database (LPD)], the Ethics Committee of the Hospital Clinic de Barcelona (Spain) reviewed and approved the protocol.

Data Sources

Details of the OMOP database sources selected for the study are shown in Supplementary Table S1. The OMOP CDM covered the specifications for all variables that were collected throughout the study using a harmonized structure and terminology system that facilitated the use of standardized analytics and tools across the network. IQVIA was the data source custodian for the selected databases and acted as the central coordinating center, providing program oversight and management, and facilitating data quality assurance. All IQVIA's databases converted to OMOP were kept secure and could not be accessed by anyone except the selected OMOP team. Standard data storage and archiving procedures were followed, as well as standard procedures to restore files in the event of a hardware or software failure.

Study Period and Population

The study period started on January 1, 2018, for all the databases, and extended to May 2022 for the Japan Claims database, September 2022 for the Italy LPD, and the UK IQVIA Medical Research Data (IMRD) and The Health Improvement Network (THIN) databases, and December 2022 for the rest of the databases. Active individuals in the databases were defined as those who had at least one clinical event (diagnosis, prescription, procedure, device, measurement, or visit to care sites) recorded in the last 12 months. For France, Germany, Italy, Spain, the UK, and Australia, the BMI thresholds applied were ≥ 25 kg/m^2 for overweight and obesity and ≥ 30 kg/m^2 for obesity alone, according to the WHO classification [15]. BMI categories based on the WHO classification were: overweight, 25.0 to < 30.0 kg/m^2; class I obesity, 30.0 to < 35.0 kg/m^2; class II obesity, 35.0 to < 40.0 kg/m^2; and class III obesity, ≥ 40 kg/m^2. For Japan, a BMI of ≥ 25 kg/m^2 was applied for obesity according to the Japan Society for the Study of Obesity (JASSO) [16]. The BMI categories based on the JASSO classification were: obesity degree 1, 25.0 to < 30.0 kg/m^2; obesity degree 2, 30.0 to < 35.0 kg/m^2; obesity degree 3, 35.0 to < 40.0 kg/m^2; and obesity degree 4, ≥ 40.0 kg/m^2. Japan has no definition for overweight according to the JASSO classification.

Adults with overweight or obesity during 2018–2022 were identified based on the presence of diagnosis codes and/or BMI as per the thresholds described above (Supplementary Table S2). Additionally, study cohorts were built to address study objectives. The prevalent cohort is reported here, which consisted of all individuals in the databases with at least one BMI measurement above ≥ 25.0 kg/m^2 in the study period (Supplementary Figure S1). The index date for this cohort was defined as the individual's highest record of BMI measurement ≥ 25.0 kg/m^2 in the study period. Included individuals were ≥ 18 years of age at index date with an observation start date ≥ 12 months before the index date. Individuals were excluded if data on age or gender were missing.

Variables and Endpoints

BMI was utilized as recorded or as derived from weight and height measurements. The BMI recording rate was estimated as the number of individuals with at least one BMI measurement among active individuals in the databases during the study period. Diagnosis codes were a combination of International Classification of Diseases (ICD) codes and others, dependent on the country (Supplementary Table S2), mapped to Systematized Nomenclature of Medicine (SNOMED) diagnosis codes in the OMOP CDM. Diagnosis codes for overweight were not available for the France LPD and Germany disease analyzer (DA). The recording rate of diagnosis codes was defined as the number of adults with at least one diagnosis code indicating overweight or obesity among the total number of individuals with overweight or obesity (based on BMI above threshold and/or diagnosis codes). As availability of waist and hip measurements was limited in the selected databases (less than 5% of patient records), these anthropometric parameters were too heterogeneous and scarce for identification of people with overweight or obesity and were not included in the analysis. ORCs of interest were pre-established in the

study protocol. Selection of ORCs of interest was guided by previous literature, clinical guidance, and the feasibility of identifying records indicating the presence of an ORC in the study datasets. The presence of an individual ORC was defined based on availability of diagnosis codes and/or medication (where possible to assign to a specific disease and available in the database), as reported within 12 months before the index date. The number of ORCs were estimated as 0, 1, 2, 3, 4, ≥ 5, ≥ 1, and ≥ 2.

Statistical Analyses

Given the descriptive nature of the study, the sample size was determined by the availability of data and the number of subjects retrieved from the databases. Analyses were conducted separately by country and database, and the General Data Protection Regulation rules were independently applied to data in each country. Analyses were conducted using OMOP analytical tools in Structured Query Language through snowflake and R (version 4.2.1; https://www.r-project.org). Descriptive statistics were reported using frequency and percentage distributions for categorical variables, including percentage of missing data as applicable. Continuous and count variables were described using mean, median, and standard deviation.

RESULTS

BMI and Diagnosis Code Recording

The number of active subjects across databases ranged from 1,203,674 in Australia to 22,413,902 in Germany (Table 1). The number of active subjects with ≥ 1 BMI recording varied widely across databases but was generally less than a quarter of the subjects in each database. The UK IMRD THIN had the highest percentage of subjects with BMI records (38.9%), followed by Spain LPD (25.0%), France LPD (22.0%), Italy LPD (21.8%), and the Japan Claims database (19.6%). The Australia EMR and the Germany DA had 4.8% and 5.5% of active subjects with ≥ 1 BMI recording, respectively.

Of the active subjects in the database with ≥ 1 BMI recording, about 60% were adults with a BMI ≥ 25 kg/m^2 in the Germany DA, Italy LPD, UK IMRD THIN, and Australia EMR, and about 40% in the France and Spain LPDs. In the Japan Claims database only 30.1% were adults with a BMI ≥ 25 kg/m^2. Of the number of adults identified with overweight or obesity (BMI and/or diagnosis code), diagnosis code recording ranged widely, from 2.9% in the France LPD to 50.1% in the Germany DA (Table 1).

BMI Distribution

The number of subjects in the prevalent cohort with a recorded BMI equal to or above the standard thresholds is shown for each database in Table 2 and in Fig. 1. In the France LPD, Germany DA, Italy LPD, Spain LPD, UK IMRD THIN, and the Australia EMR, the percentage of subjects with a BMI 25 to < 30 kg/m^2 ranged from 43.6% to 55.9% of the prevalent cohort. In Japan, this category (obesity degree 1 in the JASSO classification) included 77.9% of the prevalent cohort. For all the databases except Japan, the proportions of subjects with BMI ≥ 30.0 kg/m^2 ranged from 44.1% to 56.4%, and those with BMI ≥ 40.0 kg/m^2 from 5.0% to 10.1%. For Japan, the proportion of subjects in each BMI category dropped drastically from BMI ≥ 30.0 kg/m^2 (22.1%) to BMI ≥ 35.0 kg/m^2 (4.7%), and only 1.0% presented with a BMI ≥ 40.0 kg/m^2.

Presence of ORCs

ORCs were common in the prevalent cohort in all the databases, with most people presenting with ≥ 1 ORC (Fig. 2; Supplementary Table S3). The most frequently recorded ORCs reported in the prevalent cohort for all the databases were hypertension (range: 28.9% in the Japan Claims database to 62.3% in the Italy LPD) and dyslipidemia (range: 20.4% in the France LPD to 46.1% in the Spain LPD) (Fig. 3; Supplementary Table S4). High rates of depression were recorded in the UK IMRD THIN (27.5%) and of anxiety in the Spain LPD (28.7%). T2D ranged from 12.1% in the France LPD to 25.9% in the Spain LPD.

Table 1 BMI and diagnosis recording rates and cohort attrition (2018–2022)

	France LPD	Germany DA	Italy LPD	Spain LPD	UK IMRD THIN	Australia EMR	Japan claims
Active subjects in the databases, *n*	10,206,481	22,413,902	1,791,140	1,570,445	4,454,982	1,203,674	10,609,060
Active subjects in the database with ≥ 1 BMI recording[a], *n* (%)	2,241,019 (22.0)	1,227,731 (5.5)	389,979 (21.8)	391,947 (25.0)	1,734,788 (38.9)	57,622 (4.8)	2,082,594 (19.6)
Adults[b] identified with overweight or obesity based on BMI ≥ 25 kg/m^2, *n* (%)	873,808 (39.0)	749,242 (61.0)	222,399 (57.0)	171,981 (43.9)	1,108,481 (63.9)	33,900 (58.8)	626,808 (30.1)
Prevalent cohort[c], *n*	430,089	437,130	170,119	163,824	893,246	25,625	567,359
Adults[d] identified with overweight or obesity based on BMI ≥ 25 kg/m^2 and/or diagnosis codes, *n* (%)[e]	880,815 (8.6)	1,229,916 (5.5)	248,256 (13.9)	185,658 (11.8)	1,110,830 (24.9)	41,533 (3.5)	656,336 (6.2)
Adults[d] with a diagnosis code present[a], *n* (%)	25,251 (2.9)	616,392 (50.1)	58,430 (23.5)	67,725 (36.5)	46,769 (4.2)	10,373 (25.0)	38,327 (5.8)

BMI body mass index, *DA* disease analyzer, *EMR* electronic medical record, *IMRD* IQVIA Medical Research Data, *LPD* longitudinal patient database, *THIN* The Health Improvement Network

[a]BMI or diagnosis codes recording according to "Methods"

[b]Age ≥ 18 years at index date (date of highest BMI ≥ 25 kg/m^2 recorded during 2018–2022)

[c]Adults (≥ 18 years) with overweight or obesity (≥ 1 BMI record of ≥ 25.0 kg/m^2) and an observation start date ≥ 12 months before the index date (date of highest BMI record) during 2018–2022

[d]Age ≥ 18 years at index date (date of first BMI ≥ 25 kg/m^2 and/or diagnosis codes during 2018–2022)

[e]Percentage calculated from the total of active subjects in the database (inclusive of those aged < 18 years)

Table 2 Number of subjects with BMI at or above standard thresholds in the prevalent cohort[a], and by BMI categories

BMI (kg/m^2)	France LPD (*n* = 430,089)	Germany DA (*n* = 437,130)	Italy LPD (*n* = 170,119)	Spain LPD (*n* = 163,824)	UK IMRD THIN (*n* = 893,246)	Australia EMR (*n* = 25,625)	Japan claims (*n* = 567,359)
≥ 25.0	430,089 (100)	437,130 (100)	170,119 (100)	163,824 (100)	893,246 (100)	25,625 (100)	567,359 (100)
≥ 27.0	324,931 (75.6)	341,585 (78.1)	127,163 (74.8)	135,972 (83.0)	726,619 (81.4)	20,672 (80.7)	317,021 (55.9)
≥ 30.0	196,670 (45.7)	215,388 (49.3)	74,948 (44.1)	92,381 (56.4)	489,744 (54.8)	13,913 (54.3)	125,608 (22.1)
≥ 35.0	72,574 (16.9)	85,097 (19.5)	26,921 (15.8)	39,065 (23.9)	217,998 (24.4)	6241 (24.4)	26,451 (4.7)
25.0 to < 30.0	233,419 (54.3)	221,742 (50.7)	95,171 (55.9)	71,443 (43.6)	403,502 (45.2)	11,712 (45.7)	441,751 (77.9)
30.0 to < 35.0	124,096 (28.9)	130,291 (29.8)	48,027 (28.2)	53,316 (32.5)	271,746 (30.4)	7672 (29.9)	99,157 (17.5)
35.0 to < 40.0	48,198 (11.2)	53,561 (12.3)	18,402 (10.8)	23,788 (14.5)	128,216 (14.4)	3703 (14.5)	20,598 (3.6)
≥ 40.0	24,376 (5.7)	31,536 (7.2)	8519 (5.0)	15,277 (9.3)	89,782 (10.1)	2538 (9.9)	5853 (1.0)

Values represent *n* (%)

BMI body mass index, *DA* disease analyzer, *EMR* electronic medical record, *IMRD* IQVIA Medical Research Data, *LPD* longitudinal patient database, *THIN* The Health Improvement Network

[a]Adults (≥ 18 years) with overweight or obesity (≥ 1 BMI record of ≥ 25.0 kg/m^2) and an observation start date ≥ 12 months before the index date (date of highest BMI record) during 2018–2022

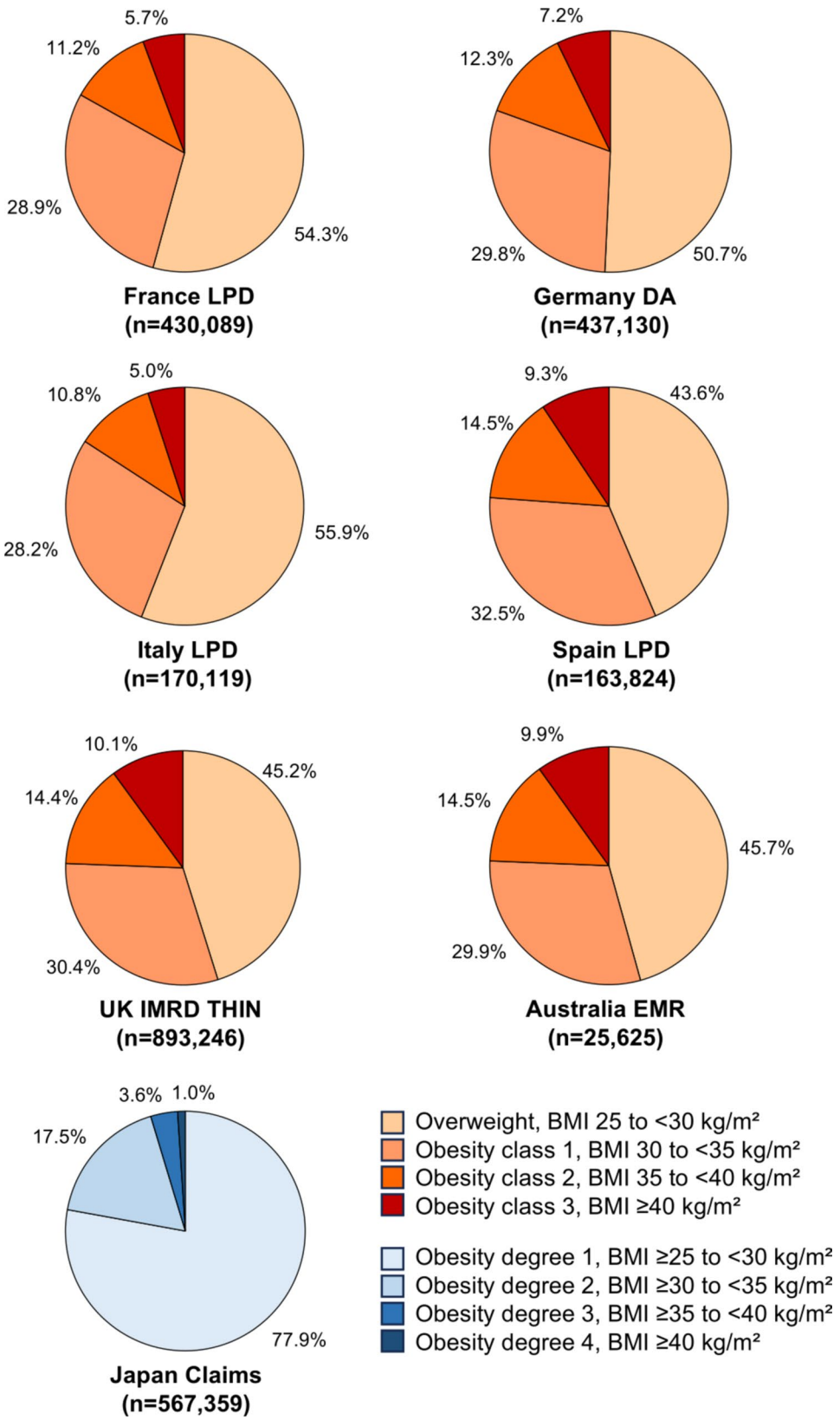
BMI and ORCs

In subjects with BMI in the range of 25 to < 30 kg/m^2, subjects with ≥ 1 ORC ranged from 56.5% in the France LPD to 82.9% in the Italy LPD (Fig. 2). In all the databases, the number of subjects with ≥ 1 ORC generally increased with increasing BMI categories, although in the Italy and Spain LPDs this tendency was not as marked as for other countries, and there was a small decrease in the number of subjects with ≥ 1 ORC in the BMI ≥ 40 kg/m^2 category. The proportion of subjects with ≥ 2 ORCs ranged from 35.9% to 68.1% (Supplementary Table S3).

DISCUSSION

The IMPACT-O study described the landscape and impact of overweight and obesity in five European and two Asia–Pacific countries for which large EMRs or claims databases were available. While data from population-based screening studies or health interview surveys suggest a high prevalence of overweight/obesity in the general population [17, 18], our results suggest that only a small proportion of people with overweight or obesity have a formally coded diagnosis and/or a BMI measurement in their records.

There are few published studies detailing the percentage of people with BMI records in healthcare databases [17, 19–23]. A systematic review of documentation of adults with overweight by UK general practitioners (GPs) showed that BMI is generally under-recorded [20]. However, a retrospective study of primary care data in the UK from 2000 to 2017 indicated that, although weight was not recorded in routine clinical practice, about one-third of patients had BMI records [22]. This is consistent with the results of our study, in which the rate of BMI recording in the UK IMRD THIN database was 38.9%. In contrast, our results that only 4.8% of people in the Australia EMR had at least one BMI record differed substantially from a published study, which reported a rate of 22.2% for patients with BMI records in GP clinics in Melbourne, Australia [19]. A survey of 10,810 people conducted over 2017–2018 in 12 European countries showed that 48.1% of participants self-reported


5.7%
11.2%
28.9%
54.3%
France LPD
(n=430,089)
7.2%
12.3%
29.8%
50.7%
Germany DA
(n=437,130)
5.0%
10.8%
28.2%
55.9%
Italy LPD
(n=170,119)
9.3%
43.6%
14.5%
32.5%
Spain LPD
(n=163,824)
10.1%
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30.4%
UK IMRD THIN
(n=893,246)
9.9%
14.5%
45.7%
29.9%
Australia EMR
(n=25,625)
3.6%
1.0%
17.5%
77.9%
Japan Claims
(n=567,359)
Overweight, BMI 25 to <30 kg/m²
Obesity class 1, BMI 30 to <35 kg/m²
Obesity class 2, BMI 35 to <40 kg/m²
Obesity class 3, BMI ≥40 kg/m²
Obesity degree 1, BMI ≥25 to <30 kg/m²
Obesity degree 2, BMI ≥30 to <35 kg/m²
Obesity degree 3, BMI ≥35 to <40 kg/m²
Obesity degree 4, BMI ≥40 kg/m²

◂**Fig. 1** Percentage of subjects in the prevalent cohort by BMI categories. The prevalent cohort included adults (≥ 18 years) with overweight or obesity (≥ 1 BMI record of ≥ 25.0 kg/m^2) and an observation start date ≥ 12 months before the index date (date of highest BMI record) during 2018–2022. *BMI* body mass index, *DA* disease analyzer, *EMR* electronic medical record, *IMRD* IQVIA Medical Research Data, *LPD* longitudinal patient database, *THIN* The Health Improvement Network

as having overweight or obesity, and 12.6% as having obesity [17]. However, the IMPACT-O study showed that only a small fraction of people could be identified with overweight or obesity based on BMI ≥ 25 kg/m^2. The results of this study suggest that, generally, BMI recording may be suboptimal.

Recording of diagnosis codes for obesity was low for most countries in the IMPACT-O study. Few studies have shown data on obesity diagnosis code recording in healthcare databases. One study showed that coding of obesity in administrative hospital discharge abstract data in Canada was poor (sensitivity, 7.75%) but accurate [24], and similar results were obtained in other studies conducted in the USA [25], and Denmark [26]. Under-recording of BMI or low use of diagnosis codes for obesity may lead to underestimation of the impact of overweight and obesity in the general population. Under-reporting of obesity has also been detected in children and adolescents [27, 28]. Diagnosis codes are the main method of health recording and statistics on disease in primary, secondary, and tertiary care, as well as on cause of death certificates. These data support payment systems, service planning, administration of quality and safety, and health services research but can also support referral and/or reimbursement of therapies in certain geographies.

It should be noted that BMI does not provide any information on body fat, nor the genetic, metabolic, physiologic, or psychologic factors associated with obesity development [29, 30]. Increasing recognition of the usefulness of measuring waist circumference for obesity-related risk assessment and the development of staging systems that assess physical, psychologic, and functional factors, such as the Edmonton Obesity Staging System [31], means that risk assessment in obesity can move beyond BMI [30]; as such, there is a need for physicians to be educated on the usefulness of these tools so that they can be incorporated into routine practice. Recently, more refined approaches including additional indicators of body fat, such as waist circumference, waist-to-hip ratio, or direct fat measurements, alongside BMI, have been proposed to improve accuracy and reduce the likelihood of misclassification [32, 33]. However, in the seven databases used in IMPACT-O, less than 5% of patients had waist circumference measurements, highlighting a critical gap between evidence-based recommendations and real-world implementation, and aligning with other reports of low recording of anthropometric parameters [34]. Additional research to further characterize the presence of ORCs across the BMI spectrum, including those with normal BMI, while also considering the effect of key confounders such as age and gender, is warranted. There is also a need to further describe the epidemiology and burden of overweight/obesity based on body composition measures.

In this retrospective real-world study, which included GP-only data for some countries and GP plus specialist data for other countries, it was not possible to investigate the cause or causes of the poor recording of obesity. However, lack of recognition of obesity as a chronic disease could potentially be the leading cause. Some countries do not formally recognize obesity as a disease [35], and GPs and specialists may record BMI and diagnosis codes differently; it has been suggested that GPs are not adequately trained to recognize overweight and obesity [36, 37]. Lack of training to manage obesity as a disease is one of the most commonly reported barriers to addressing obesity management by primary care physicians [38], and lack of specificity and low sensitivity in the BMI-related diagnosis codes has been reported [39]. It is also possible that GPs do not record overweight or obesity unless there is a clear impact on the health of patients, especially where limited human and financial resources are available in primary care services [40]. Primary care physicians may perceive obesity in the context of other chronic diseases. Prior studies have shown that weight could be

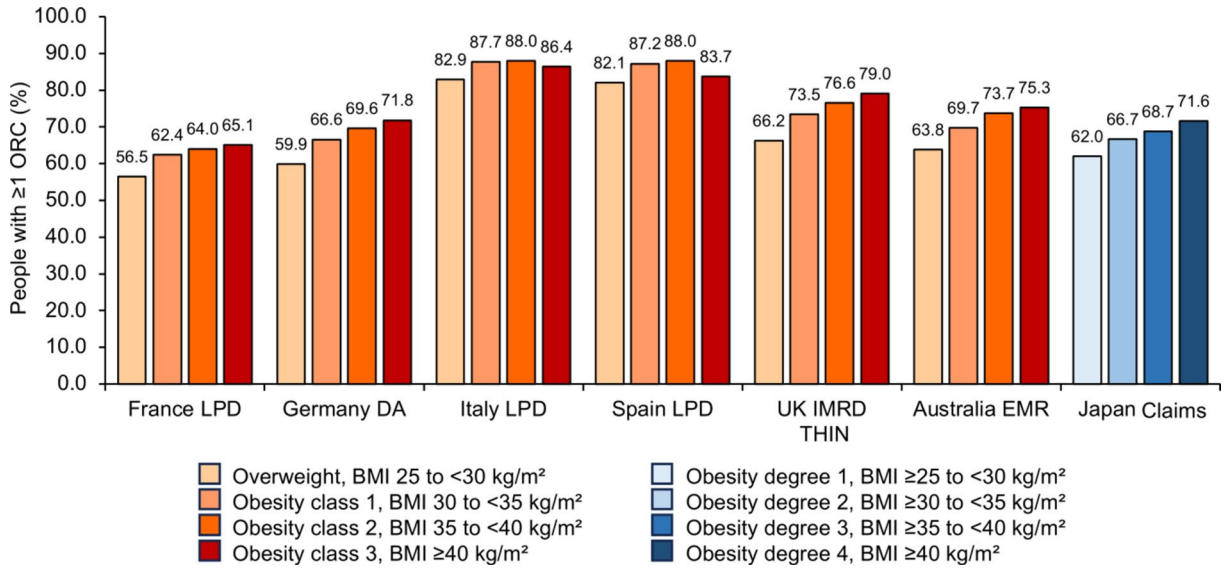

Fig. 2 Percentage of subjects in the prevalent cohort with at least one ORC, by BMI categories. The prevalent cohort included adults (≥ 18 years) with overweight or obesity (≥ 1 BMI record of ≥ 25.0 kg/m^2) and an observation start date ≥ 12 months before the index date (date of highest BMI record) during 2018–2022. *BMI* body mass index, *DA* disease analyzer, *EMR* electronic medical record, *IMRD* IQVIA Medical Research Data, *LPD* longitudinal patient database, *ORC* obesity-related complication, *THIN* The Health Improvement Network

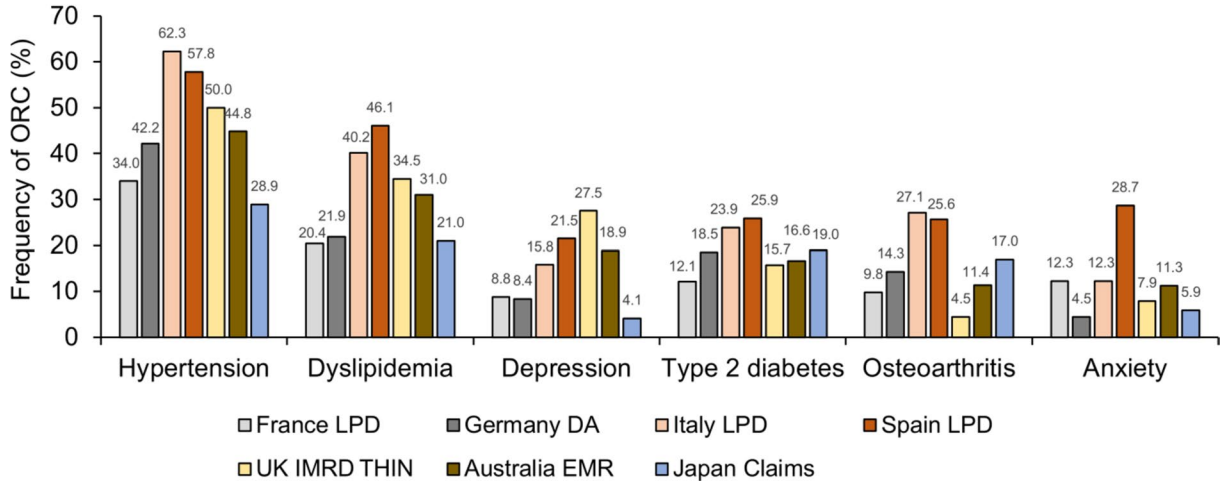

Fig. 3 ORCs occurring in ≥ 25% of the prevalent cohort in at least one country. The prevalent cohort included adults (≥ 18 years) with overweight or obesity (≥ 1 BMI record of ≥ 25.0 kg/m^2) and an observation start date ≥ 12 months before the index date (date of highest BMI record) during 2018–2022. See Supplementary Table S4 for a full list of ORCs. Hypertension, dyslipidemia, depression, T2D, and anxiety were identified by diagnosis code and/or medication. Osteoarthritis was specified by diagnosis code only. *DA* disease analyzer, *EMR* electronic medical record, *IMRD* IQVIA Medical Research Data, *LPD* longitudinal patient database, *ORC* obesity-related complications, *T2D* type 2 diabetes

recorded preferentially and more frequently for people with chronic conditions, for whom regular weight measurement is recommended in clinical guidelines, and for which weight recording is a performance indicator [19, 21, 41]. In the Netherlands, 90% of people with diabetes had at least one weight measurement recorded in their EMR, but only 33% of people without any ORCs had a record [21]. The presence of ORCs is cited in some studies as the main reason for BMI recording and weight interventions by healthcare professionals [41, 42]. Consistent with this possibility, the present study found that most people in the database had at least one ORC. Also, age could be a factor that affects frequency of BMI recording, as people with chronic conditions tend to be of advanced age and older age could also lead to increased BMI recording [21]. Prior studies suggest that BMI recording is captured most likely among people with the highest BMI [41]. A study in the USA suggested that individuals with class II and III obesity (BMI ≥ 35 kg/m^2) were more likely to have a diagnosis coded in the claims database, but not those with a BMI < 35 kg/m^2 [25]. Consistent with this, the results of the IMPACT-O study showed that for Spain, the UK, and Australia, most people with at least one BMI record had a BMI > 30 kg/m^2.

While the consequences of suboptimal coding of overweight or obesity in healthcare databases are beyond the scope of the IMPACT-O study, prior studies have shown that underestimation of the impact of overweight and obesity in healthcare settings could potentially lead to lower quality of care, as people with overweight or obesity have an increased risk of developing ORCs [43, 44]. A study in the UK concluded that underestimation of obesity leads to missed opportunities to provide advice to address the disease in primary care [45]. A study in Australia indicated that low recording also has a financial impact with poor obesity recording resulting in lost funding opportunities and increased burden to hospitals [46]. Given the poor recording of obesity-related ICD codes in claims databases, other approaches should be considered to assess the incidence of obesity. For example, claims-based models have been developed to identify obesity categories in claims databases, when BMI measurements are missing [47], and population-based screening or health interviews for self-reported obesity could also be conducted to support these research questions [48]. Generally, including diagnosis of obesity in the medical records would likely improve the chances of starting a treatment, as suggested by a study in the USA, which showed that the only factor that increased the probability of receiving treatment for obesity was having been diagnosed with overweight or obesity in the medical record [49]. Having BMI records in databases could also provide valuable information on the efficacy of weight management strategies over time. Medical record databases should also be used to identify the frequency of medically recognized overweight or obesity, and to assess changes over time with increased recognition of obesity as a chronic disease [1]. Obesity stigma, or the lack of therapeutic options when this study was conducted, could have contributed to the suboptimal recording of obesity detected in this study. Further insights into the barriers and drivers for BMI and diagnosis code recording across geographies are needed.

The relationship between obesity and multimorbidity has been described before [50, 51]. In our study, most people (60–85%) in the prevalent cohort presented at least one ORC across countries and more than 50% of people with BMI of 25 to < 30 kg/m^2 had at least one ORC. A large multinational survey has shown that both the rate and burden of multimorbidity generally increases with increasing BMI [3]. The association of obesity with increased multimorbidity has been observed in older adults in a study in the USA [52]. In IMPACT-O, the burden of ORCs generally increased with increasing BMI, except in Spain and Italy, which presented the highest proportion of patients with at least one ORC, whatever the obesity class. It should be noted that, even in the range of BMI 25 to < 30 kg/m^2, the burden of ORCs was substantial, suggesting that BMI might not be the only factor associated with ORCs.

Obesity has been associated with at least 21 cardiometabolic, digestive, respiratory, neurologic, musculoskeletal, and infectious diseases [51]. Our results showed that hypertension and dyslipidemia were consistently the most

frequent ORCs across countries, similar to the results of prior studies [41]. The RESOURCE survey of six European countries showed that the most frequently reported ORCs were hypertension (39.3% of participants), dyslipidemia (22.8%), and T2D (17.5%) [53]. In our study, a high frequency of mental health complications was also found (depression and anxiety). Numerous studies have described obesity as a risk factor for mental health issues [54], and an association between obesity, reduced HRQoL, and self-esteem has also been established [10]. In contrast, for certain conditions which are generally underdiagnosed in routine care, such as obstructive sleep apnea, the IMPACT-O study showed low recording rates [55, 56].

A strength of the IMPACT-O study is its large sample sizes and the fact that it represents routine clinical practice in the way obesity is recorded, coded, and managed. IMPACT-O also has some limitations that should be taken into account when interpreting the data. First, EMR data sources are not traditionally collected for research purposes but for continued patient care, and therefore can be incomplete or result in selection bias. For example, as in previous reports, waist and hip measurements were not available for most patients and could not be used in the analysis. Also, time since diagnosis of obesity could not be reported, and only diagnosis records and/or BMI measurement at visits within the study period were considered. Second, there is the possibility of misclassification of diagnosis and/or BMI categories and under-recording. For example, healthy adults with elevated BMI but no complications may not be captured in these databases if they have no history of healthcare visits, particularly in claims databases where encounters are driven by medical necessity. This selection bias is an inherent limitation of real-world data studies and may lead to overestimation of complication rates, especially in lower BMI categories. Also, some conditions may be poorly recorded in the existing databases, and this might lead to misclassification of people. This particularly affects data on the frequency of certain ORCs, resulting in underestimation when compared to other studies using primary data collection methods (e.g., surveys). Third, data were derived from individuals who visited healthcare providers and healthcare centers participating in the database panels. While findings are expected to reflect real-world and routine management practices in each country, they may also be affected by the characteristics of the databases, which are different in each country; for example, some databases included only GP records, and others included data from GPs and specialists. Therefore, caution should be exercised when interpreting results. Direct comparisons between countries should be avoided, as the study was not intended to draw comparisons but rather to get an overall interpretation of the landscape across different geographies based on large datasets.

CONCLUSION

While the prevalence of overweight and obesity has been estimated to be high in population-based screening studies, the results from the IMPACT-O study suggest that only a small proportion of people with overweight/obesity have a formal, documented, coded diagnosis in their healthcare records. Also, many people have missing BMI records, suggesting that obesity and overweight might be globally underestimated in healthcare databases. The IMPACT-O study showed that most adults with a record of BMI ≥ 30 kg/m^2 in the databases had more than one ORC and, even more notably, that > 50% of people with BMI 25 to < 30 kg/m^2 present with at least one ORC, suggesting that there is a need for earlier and more intensive therapies for the treatment of adults with overweight or obesity. In addition, the results suggest that the recording of overweight or obesity may be more systematic in patients with higher burden, and/or that BMI might not be the only factor associated with ORCs, and other parameters such as waist circumference, waist-to-hip ratio, or body composition might be more relevant to assess the risk of complications for these patients. Further studies are needed to understand the underlying reasons for the observed rates of overweight and obesity recording in the IMPACT-O study. Diagnosing obesity is the first step towards effective treatment, as under-recording of obesity in

databases may lead to an underestimation of the impact of obesity on healthcare settings and, as a result, to a lower quality of care and missed opportunities to provide early and effective medical support for obesity management.

Overall, despite cross-country heterogeneity of data sources and variability in clinical documentation practice, the findings from the IMPACT-O study consistently suggest that recording of obesity in healthcare settings is insufficient worldwide. Strengthening efforts via coordinated national public health interventions to increase awareness of overweight and obesity and recording levels in healthcare settings is warranted. Routine measurement and recording of parameters that aid the diagnosis, monitoring, and stratification of overweight/obesity (e.g., BMI, adiposity measures, etc.) according to latest clinical guidelines and recommendations is needed.

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Data Availability. All data relevant to the study are included in the article or uploaded as supplementary information.

Declarations

Conflict of Interest. Esther Artime and Sarah Zimner-Rapuch are employees and minor shareholders of Lilly and Company. Anastasia Lampropoulou was an employee and minor shareholder of Eli Lilly and Company at the time the work was conducted and is a current employee and minor shareholder of Novartis. Erik Spaepen is an external consultant of Eli Lilly and Company. Carel Le Roux has received consulting fees for advisory board participation from Eli Lilly, NovoNordisk, Johnson & Johnson, Boehringer Ingelheim, GI Dynamics, Herbalife, Altimmune, Irish Life Health, Roche and AstraZeneca; speaker fees from NovoNordisk, Herbalife, Johnson & Johnson, Eli Lilly, Boehringer Ingelheim, Rhythm Pharmaceuticals and Currax Pharmaceuticals; support for travel to/attending meetings from NovoNordisk, Herbalife, Johnson & Johnson, Eli Lilly and Boehringer Ingelheim; institutional grants from the Irish Research Council, Health Research Board, Science Foundation Ireland, and Anabio; and holds stock or stock options in Keyron and Beyond BMI. Dror Dicker has received royalties or licenses, consulting fees, speaker fees, and support for travel to/attending meetings from NovoNordisk, Eli Lilly

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Ethical Approval. All analyses performed in this study were conducted in accordance with Data Use Agreement terms as specified by the data owners. For the databases for France, Germany, Italy, the UK, Australia, and Japan, there was no Institutional Review Board applicable to the usage and dissemination of the results of this study or required registration of the protocol with additional ethics oversight. For the database for Spain LPD, the Ethics Committee of the Hospital Clinic de Barcelona (Spain) reviewed and approved the protocol.

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